Assignment 8:

Write a C program that includes a user-defined function named countSetBits with the signature int countSetBits(int num);. The function should count and return the number of set bits (1s) in the binary representation of the given number.

Source Code:

```
#include<stdio.h>
int countSetBits(int num) {
   int count = 0;
   while (num>0) {
      if (num & 1)
         count++;
      num = num >> 1;
   }
   return count;
}
int main() {
   int num, count;
   printf("Enter the number : ");
   scanf("%d",&num);
   count = countSetBits(num);
   printf("The number of set bits = %d",count);
   return 0;
}
```

Output:

```
Enter the number :5

The number of set bits = 2
Process finished with exit code 0

Enter the number :13

The number of set bits = 3
Process finished with exit code 0
```